b) It has a composer playground, which allows users to run composers along with setting up the local blockchain network. You can read more about this at *https://hyperledger.github.io/.composer/latest/*.

Figure 5 shows an abstract layer over Fabric that lets you make a blockchain model and run it on a browser.

**Testnet:** This can quickly fire up a personal blockchain which can be used to run tests, execute commands and inspect the state while controlling the operation of the chain.

**Gnache:** Previously known as TestRPC, Gnache provides a local blockchain for development purposes. It comes with an interface with which you can see and interact with your private blockchain. It provides information about the transaction count, balances, etc.

***Features***

a) We can run tests and know about the internal state of the network using Gnache.
b) It has the output log and other internal blockchain information, including responses and other debugging information. More can be read at *https://truffleframework.com/gnache*.

## Frontends

**Drizzle:** This is a Front-ends library collection that is used for making dApps user interfaces. It comes with a predefined configuration, which takes care of synchronising smart contract data. Also, it comes with React components for commonly used functions like generating an input form for contracts. Further details can be read at *https://truffleframework.com/drizzle*.

***Features***

a) Its architecture is completely modular and can be modified according to the need.
b) As of now Drizzle-React is available for direct use but making it Drizzle-React is not difficult because of the architecture.

Figure 7: Drizzle – A modular front-end module configured to interact with smart contracts.

Figure 8: Decentralised hash based storage

Figure 9: CouchDB

**Docker images:** All major blockchain frameworks have their Docker image available at *hub.docker.com*.

## Storage

**IPFS (Interplanetary File System)**: As the name suggests, it can help you store and create Web3 storage. It uses decentralised storage and links every file using a hash instead of an address reference as is currently used by our standard Internet system. Hashes generated can be stored into the blockchain instead of storing the whole file. It can be used with any blockchain network. A detailed explanation can be found at *http://www.ipfs.io*.

**CouchDB:** This is a key-value state database embedded in the peer process. It is an alternative to levelDB (also, a key pair database) for a state database in Hyperledger Fabric. Additionally, it lets you query state database in JSON format, querying by a range or composite keys for multiple parameters. It syncs data between server clusters and mobile phones, Web apps and others, enabling a complete offline-first user experience. CouchDB is internally fault-tolerant and a single problem does not cascade to affect the entire database, but remains in an isolated request.

## BaaS: Blockchain-as-a-Service

As blockchain has gone mainstream, tech giants have released their own versions of it as a service. We can add the needed members and everything else is handled by the service itself. These services are flexible and can be integrated with existing systems using REST APIs.

**Microsoft Azure:** This created the so called BaaS or Blockchain-as-a-Service, a useful tool for developers to build dApps in a safe and cheaper environment — one that supports several chains, including MultiChain, Eris, Storj, and Augur. BaaS was created mainly to enable the backend capabilities needed by blockchain solutions. On Azure, a blockchain can be launched with just a few clicks, rather than by setting up your own. More information is available at *https://azure.microsoft.com/en-in/features/blockchain-workbench/*.

**AWS:** This is a template provided by Amazon Web Services to create and deploy blockchain networks very fast and in an easy manner (*https://aws.amazon.com/blockchain/*) END

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